Villa Composition & Budget Logic

AID-LOG-008 | Revision 1

Purpose. Defines the project-level backend logic used by AIDOOi to coordinate multiple room modules across one or more villa floors, maintain whole-villa design coherence, allocate a total project budget, and generate a unified villa recommendation.

> **Core principle: Villa Logic coordinates multiple room modules across multiple floors; it does not recreate the individual room composition logic.**

1. Villa Project Architecture

A Villa is treated as one coordinated multi-room and potentially multi-floor furnishing project. Each room module is resolved through its applicable room-level LOG standard before project-level coordination is applied.

Room Module	Referenced Logic	Application
Each Bedroom	AID-LOG-003	Repeated for every bedroom and retained with its floor assignment.
Each Living Area	AID-LOG-002	Repeated for every living area and retained with its floor assignment.
Each Dining Area	AID-LOG-004	Repeated for every dining area and retained with its floor assignment.
Home Office	AID-LOG-005	Included only when selected by the customer.

> **Project sequence: Number of Floors -> Room Modules + Floor Assignment -> Individual Room Fit & Composition -> Repeated-Space Preference -> Cross-Room Coordination -> Cross-Floor Coordination -> Villa Project Composition.**

2. Cross-Room & Cross-Floor Coordination

Overall Style is the primary design anchor for the entire villa. Individual rooms and floors may express different functional character while remaining within the selected style direction.

Overall Colour Direction defines a coordinated palette across the villa. It does not require identical product colours in every room or floor.

Floor Character Allowed. Different floors may have distinct emphasis - for example, a more formal ground floor and a calmer upper floor - while remaining visually related to the same villa project.

> **Whole-villa coherence should be maintained without forcing identical design across rooms or floors.**

Cross-project controls: Style Coherence + Colour Harmony + Material Compatibility + Overall Budget Balance.

3. Repeated-Space Consistency Preference

A consistency question is shown only when two or more spaces of the same type exist. This keeps the customer journey lean while allowing meaningful control over repeated rooms.

Condition	Customer Preference	Logic
2+ Bedrooms	Same Products / Different Products	Apply across bedrooms where dimensions, bed size, fit and availability allow.
2+ Living Areas	Same Products / Different Products	Apply across living areas where dimensions, composition and fit allow.

Condition	Customer Preference	Logic
2+ Dining Areas	Same Products / Different Products	Apply across dining areas where capacity, dimensions and fit allow.

Same Products is a preference, not a fit override. Room dimensions, circulation, required capacity and functional requirements always take priority.

Same Products does not create an assumed quantity discount. Seller discounts are applied only when present in the actual offer.

4. Composition Principles

1. Each room retains its own room-level composition and fit logic.

2. Additional rooms and floors increase project scope; additional floor area alone must not create unnecessary furniture.

3. Project coordination must not override functional completeness or circulation within an individual room.

4. Repeated rooms may use the same or different products according to customer preference, subject to fit and availability.

5. Larger project scope should improve coordination and quality rather than create artificial product duplication.

5. Villa Budget Architecture

Villa budgeting is module-based. Unlike Full Apartment logic, the villa does not use a fixed total-budget table based only on bedroom count because the number of Living Areas, Dining Areas and other included room modules can vary substantially.

> **Total Villa Budget = Sum of Bedroom Modules + Living Modules + Dining Modules + Home Office Module (if selected).**

Villa Module	Budget Source	Budget Basis
Each Bedroom	AID-LOG-003	Actual room dimensions, bed size and composition.
Each Living Area	AID-LOG-002	Actual room dimensions, size classification and composition.
Each Dining Area	AID-LOG-004	Actual room dimensions, dining capacity and composition.
Home Office	AID-LOG-005	Actual room dimensions, workspace users and composition.

The customer still selects one total Villa Project Budget using the shared tiers: Essential, Value, Standard, Premium, Luxury. Backend allocation may mix room-level and product-level tiers to optimize the complete villa.

> **Villa Budget is driven by actual room modules and their composition requirements, not by bedroom count or number of floors alone.**

Number of Floors is not an independent budget modifier. Furnishing scope is determined by the actual spaces included in the project.

6. Budget Allocation & Constraint Logic

> **Allocation priority: Required Room Modules -> Essential Furniture within each Room -> Fit & Function -> Product Tier Optimization -> Secondary Furniture -> Decor.**

Functional Completeness Rule: Every required room module must first achieve functional completeness before additional budget is used to upgrade other rooms.

No room receives automatic budget priority based solely on sequence or floor. Allocation is driven by actual dimensions, composition and functional requirements.

If budget is constrained: Preserve Fit -> Preserve Essential Products -> Lower Product Tiers -> Remove Conditional / Secondary Products -> Reduce Decor -> Surface Budget Constraint.

If budget exceeds essential composition needs: Improve Product Quality -> Improve Product Scale / Fit -> Improve Material / Design Match -> Enhance Secondary Furniture -> Enhance Decor.

AIDOOi must not add unnecessary products merely to consume available budget.

7. Final Villa Recommendation Logic

> **Villa Structure -> Floor Assignment -> Room Modules -> Individual Room Fit & Composition -> Repeated-Space Preference -> Cross-Room & Cross-Floor Coordination -> Total Villa Budget -> Dynamic Budget Allocation -> Product Selection -> Whole-Villa Recommendation.**

Rule	Requirement
Whole-Project Recommendation	Present the villa as one coordinated project rather than unrelated room recommendations.
Room Integrity	Each room must continue to satisfy its applicable room-level LOG standard.
Repeated-Space Preference	Apply Same / Different Products preference without overriding fit, capacity or function.
Budget Balance	No room should lose functional completeness solely to make another room more premium.
Design Coherence	Maintain one overall design direction while allowing room and floor character.
Dynamic Recalculation	Recalculate Total Villa Price and budget balance when products are removed or substituted.

Product selection priority: Fit & Function -> Availability -> Total Project Budget -> Overall Style -> Colour Harmony -> Material Compatibility -> Product Quality / Value.

Final principle: AIDOOi treats the villa as one coordinated multi-floor project, while preserving the functional identity, fit requirements and design character of every individual room.

8. Standard References

Reference	Standard
AID-SPC-008	Villa Path Standard
AID-LOG-001	Product Price Range Standard
AID-LOG-002	Living Room Composition, Fit & Budget Logic
AID-LOG-003	Bedroom Composition, Fit & Budget Logic
AID-LOG-004	Dining Room Composition, Fit & Budget Logic
AID-LOG-005	Home Office Composition, Fit & Budget Logic
AID-DATA-007	Product Colour Standard
AID-DATA-009	Product Style Standard

Revision 1 status: Approved MVP project logic. Module weighting, allocation behavior and coordination rules may be refined later using observed inventory, customer behavior and recommendation performance.